# Do Now · retrieval

## Lesson 078 · Binary Search: Bounds and Sorted Data

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Binary search precondition?

### Q2

What does low represent?

### Q3

What is mid for bounds low0 high6?

### Q4

Binary search target15 in [1,3,5,7,9,11,13,15]: give midpoint indexes.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Understand — Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

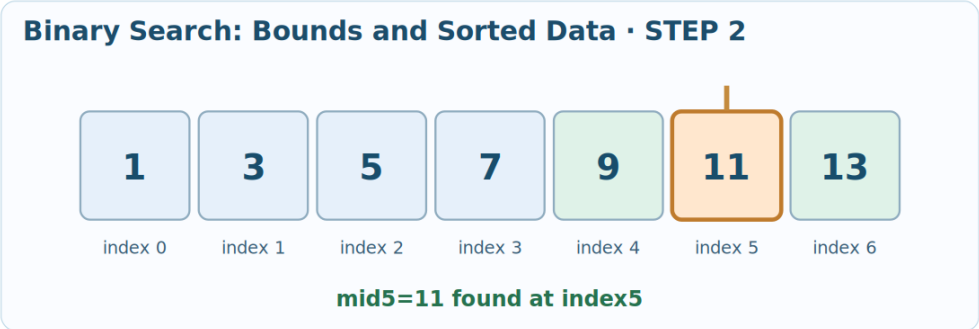
## Core idea 2

Trace — At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

## Core idea 3

Explain — Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

low — Lowest candidate index.

high — Highest candidate index.

midpoint — Middle candidate.

halving — Repeatedly reducing interval by about half.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

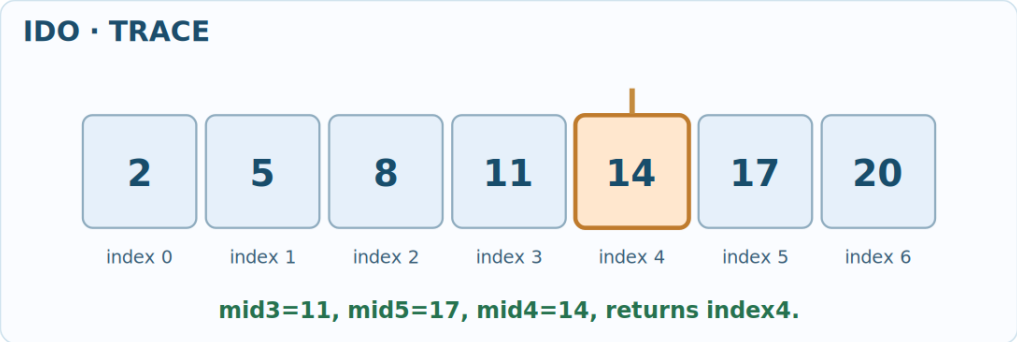
## Problem

Find14 in [2,5,8,11,14,17,20].

## Follow the worked method

1. Check the working — mid3=11, mid5=17, mid4=14, returns index4.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Check the working
mid2=5, mid0=1, mid1=3, low2>high1: return−1.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Trace target11 in [1,3,5,7,9,11,13]. [6]

## Question 2

Write three binary-search comparison cases and correct bounds. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Compare iterative binary and linear search with respect to time and precondition. [6]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

After a too-small midpoint, setting low=mid always ensures progress.

## Use key vocabulary from memory

1. Explain low in your own words.

2. Explain high in your own words.

3. Explain midpoint in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.